

ISYS1055/3412 (Practical) Database Concepts

Assessment 4: Database Design Project

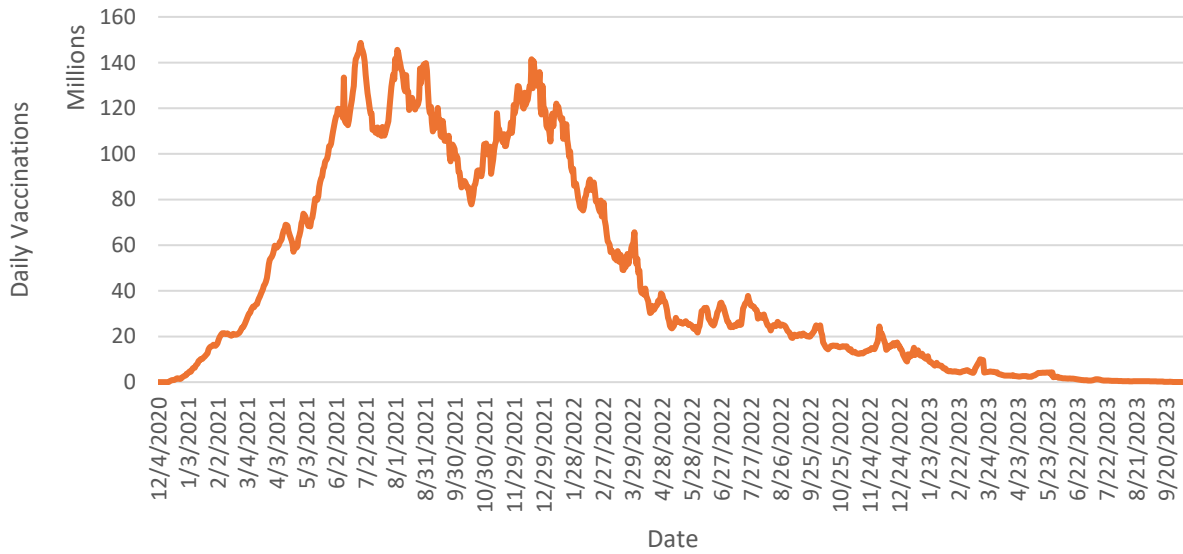
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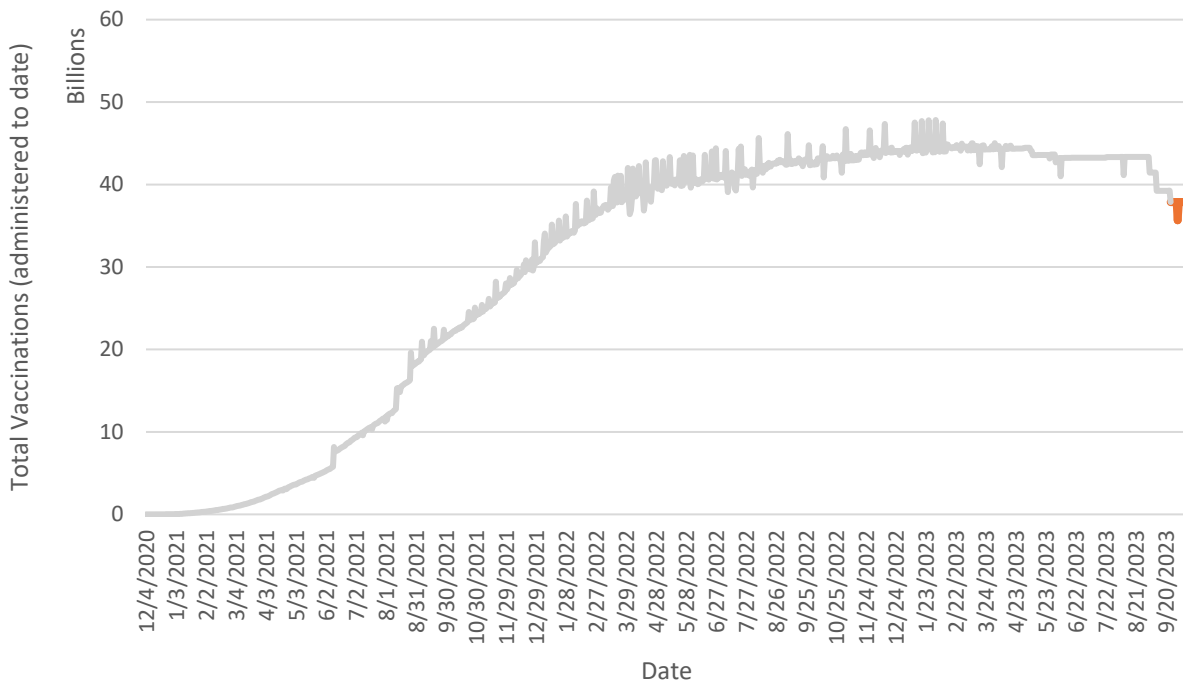
Problem D1:

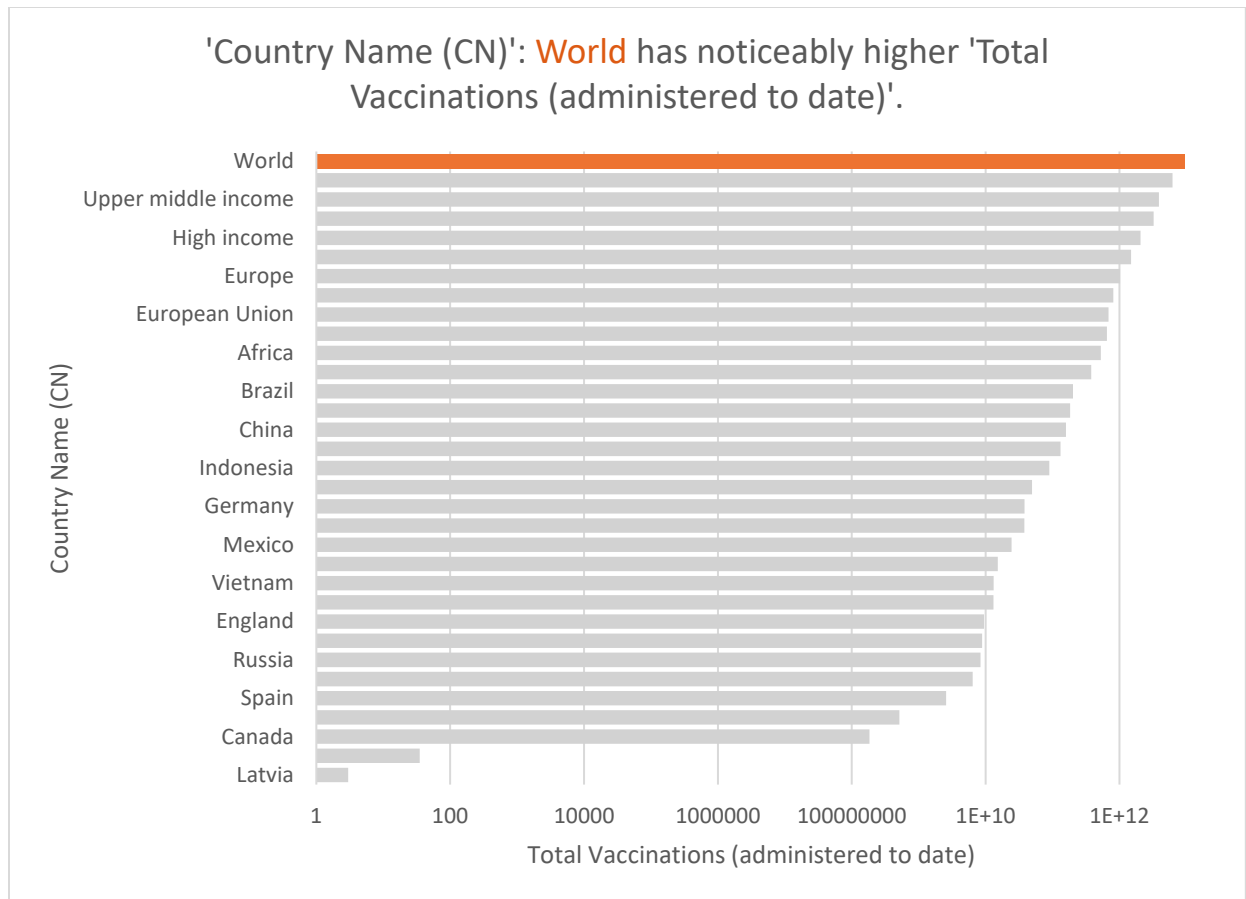
```
SELECT c.country AS 'Country Name (CN)',
       (v.peopleVaccinated + v.peopleFullyVaccinated
        + v.totalBoosters) AS 'Total Vaccinations (administered to date)',
       v.dailyVaccinations AS 'Daily Vaccinations',
       v.dates AS 'Date'
FROM vaccinations v
JOIN country c
ON v.iso = c.iso
JOIN (
    SELECT dates, AVG(peopleVaccinated + peopleFullyVaccinated
                      + totalBoosters) AS avg_vaccinations
    FROM vaccinations
    GROUP BY dates
) av ON v.dates = av.dates
WHERE (v.peopleVaccinated + v.peopleFullyVaccinated
       + v.totalBoosters) > av.avg_vaccinations;
```

Daily Vaccinations shows a repeating pattern.



'Total Vaccinations (administered to date)' changed significantly from 'Date': 9/25/2023 to 10/12/2023.





Problem D2:

```

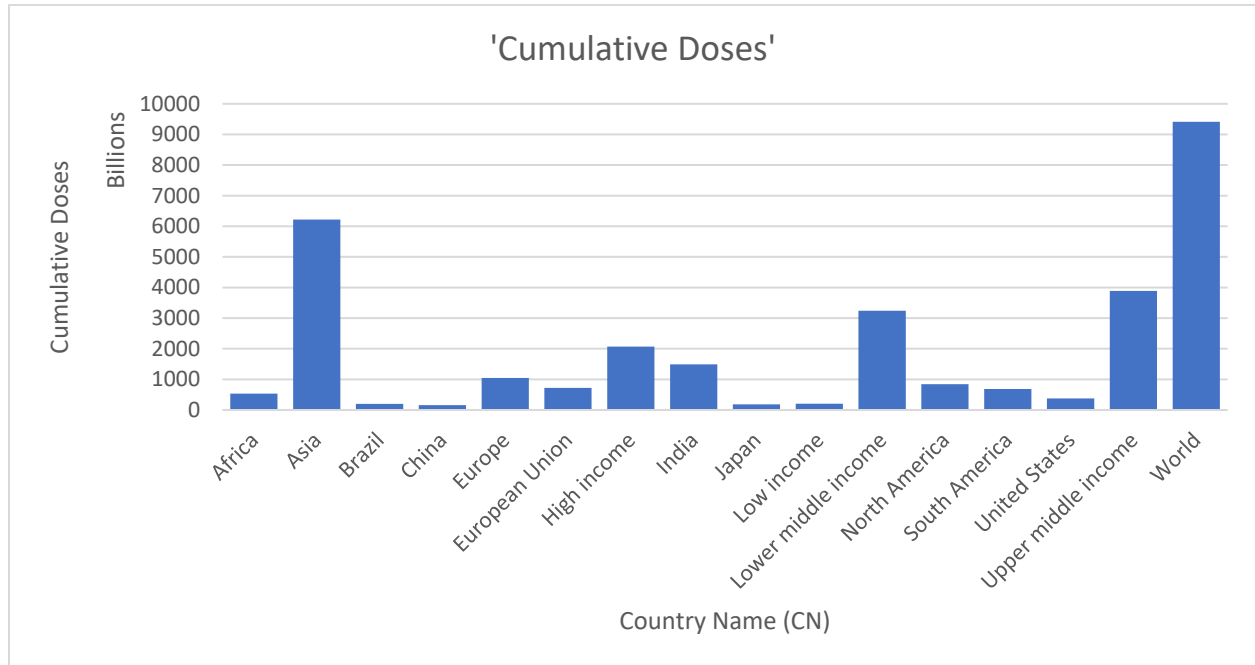
SELECT c.country AS 'Country Name (CN)',
       (SUM(v.peopleVaccinated
+ v.peopleFullyVaccinated
+ v.totalBoosters))
       AS 'Cumulative Doses'
FROM vaccinations v
JOIN country c
ON v.iso = c.iso
GROUP BY c.country
HAVING (SUM(v.peopleVaccinated + v.peopleFullyVaccinated + v.totalBoosters)) > (
  SELECT AVG(total_vaccinations)
  FROM (
    SELECT c.country, (SUM(v.peopleVaccinated
+ v.peopleFullyVaccinated + v.totalBoosters)) AS total_vaccinations
    FROM vaccinations v
    JOIN country c
    ON v.iso = c.iso
  )
)

```

```

GROUP BY c.country
)
);

```

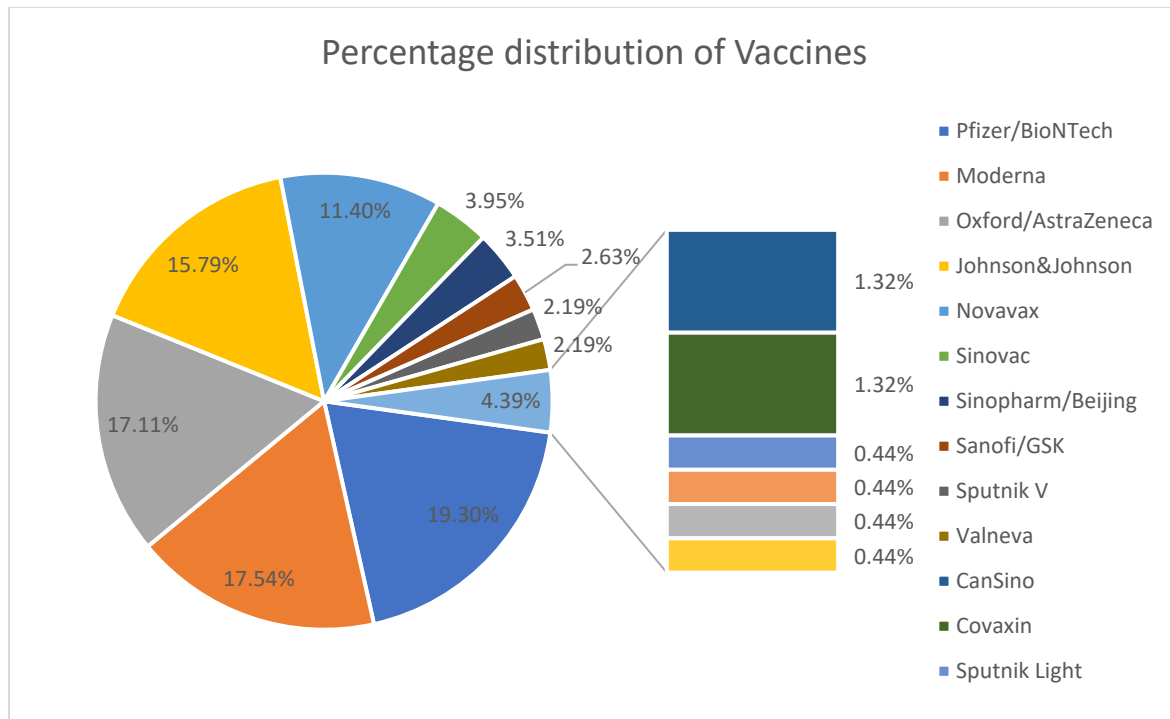


Problem D3:

```

SELECT DISTINCT c.country AS 'Country',
               vm.vaccines AS 'Vaccine Type'
FROM vaccinationsByManufacturer vm
JOIN country c ON vm.iso = c.iso;

```



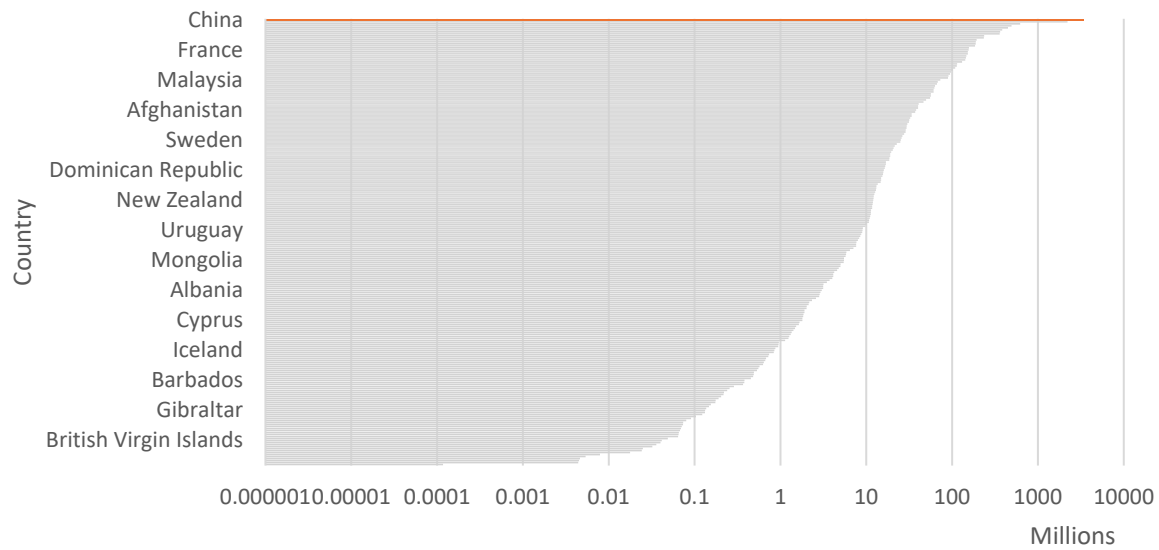
Problem D4:

```

SELECT c.country AS 'Country',
       s.sourceName AS 'Source Name (URL)',
       MAX(v.peopleVaccinated + v.peopleFullyVaccinated
          + v.totalBoosters) AS 'Biggest total Administered Vaccines'
FROM vaccinations v
JOIN country c ON v.iso = c.iso
JOIN dataFrom d ON v.iso = d.iso
JOIN sources s ON d.url = s.url
GROUP BY c.country, s.sourceName
ORDER BY s.sourceName;

```

Biggest total Administered Vaccines by Country



Biggest total Administered Vaccines